

LAB 3: Propositional Logic & Model Checking

Group Challenge

Instructor	Dr. Naila Marir	Duration	1 hour 30 minutes
Semester	Fall 2026	Group size	3–4 students
Submission	One GitHub link per group	Total marks	100
Group No.			
Member name	Student ID	Member name	Student ID
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1. Objectives

1. Represent facts as propositional logic sentences using Not, And, Or and Implication.
2. Explain how `model_check` decides whether a knowledge base entails a query.
3. Encode a new logic puzzle in Python and solve it with model checking.

2. Tools and Files

1. Python 3 and the package `termcolor` (install with `pip install termcolor`).
2. Lab files (on Blackboard): `logic.py`, `harry.py`, `clue.py`, `puzzle.py`, `mastermind.py`. Put all of them in one folder.
3. A GitHub account for each group.

3. Lab Plan

Step	Activity	Time
0	Setup: run <code>harry.py</code> and <code>clue.py</code>	5 min
1	Task 1 — Understand the code	20 min
2	Task 2 — Change the knowledge	20 min
3	Task 3 — Knights & Knaves challenge	30 min
4	Write the PDF and push to GitHub	15 min

4. Task 1 — Understand the Code (30 marks, 20 min)

1. **harry.py:** Write the truth table for `rain`, `hagrid`, `dumbledore` (8 rows). Mark the rows where the knowledge base is true. Use the table to explain why the program prints `True`.

$KB = (\neg \text{rain} \Rightarrow \text{hagrid}) \wedge (\text{hagrid} \vee \text{dumbledore}) \wedge \neg(\text{hagrid} \wedge \text{dumbledore}) \wedge \text{dumbledore}$

Rain	Hagrid	Dumbledore	$\neg \text{rain} \Rightarrow \text{hagrid}$	$h \vee d$	$\neg(h \wedge d)$	Dumbledore	KB
T	T	T	T	T	F	T	F
T	T	F	T	T	T	F	F
T	F	T	T	T	T	T	True
T	F	F	T	F	T	F	F
F	T	T	T	T	F	T	F
F	T	F	T	T	T	F	F
F	F	T	F	T	T	T	F
F	F	F	F	F	T	F	F

The KB is true in only one row (rain = T, hagrid = F, dumbledore = T). `model_check(knowledge, rain)` returns True only if the query is true in every model where the KB is true. There is just one such model and rain is true in it, so it prints True.

```
● amratiah@MacBook-Air-3 CS3081 % /usr/local/bin/python3 /Users/amratiah/CS3081/Labs/Lab2/harry.py
True
```

2. **clue.py:** The program prints MsScarlet, library and knife. Explain in plain English why each one must be true.

- MsScarlet: one of Mustard, Plum or Scarlet did it. Mustard and Plum are ruled out by known cards, so it must be Scarlet.
- library: the room is the ballroom, kitchen or library. The kitchen and ballroom are ruled out, so it is the library.
- knife: the weapon is the knife, revolver or wrench. The revolver is ruled out. The unknown card says not all of Scarlet, library and wrench are true together. Scarlet and library are already true, so the wrench must be false. That leaves the knife.

The program prints these in the order of the symbols list: MsScarlet, library, knife. The other symbols are definitely false, so they print nothing.

```
● amratiah@MacBook-Air-3 CS3081 % /usr/local/bin/python3 /Users/amratiah/CS3081/Labs/Lab2/clue.py
MsScarlet: YES
library: YES
knife: YES
```

5. Task 2 — Change the Knowledge (30 marks, 20 min)

For each change: **(a)** predict the result, **(b)** run the code, **(c)** explain the result.

1. **harry.py:** Delete the last fact `dumbledore` from the knowledge base. What does the program print now? Why?

a Prediction: it prints False.

b

```
Labs > Lab2 >  harry.py > ...
1  from logic import *
2
3  rain = Symbol("rain")
4  hagrid = Symbol("hagrid")
5  dumbledore = Symbol("dumbledore")
6
7  knowledge = And(
8      Implication(Not(rain), hagrid),
9      Or(hagrid, dumbledore),
10     Not(And(hagrid, dumbledore))
11 )
12
13 print(model_check(knowledge, rain))
14
```

```
● amratiah@MacBook-Air-3 CS3081 % /usr/local/bin/python3 /Users/amratiah/CS3081/Labs/Lab2/harry.py
False
```

c Why: the KB now only says exactly one of hagrid and dumbledore is true. In the model (`rain = F`, `hagrid = T`, `dumbledore = F`) the KB is still true, but rain is false. Rain is therefore not entailed, and `model_check` returns False.

2. **puzzle.py:** Delete the clue `MinervaGryffindor`. Who can still be placed in a house? Then add **one new clue of your own** that makes the answer unique again.

a Prediction: nobody can be placed, so nothing prints.

b

```
● amratiah@MacBook-Air-3 CS3081 % /usr/local/bin/python3 /Users/amratiah/CS3081/Labs/Lab2/puzzle.py
○ amratiah@MacBook-Air-3 CS3081 %
```

c Why: the remaining clues are only that Gilderoy is in Gryffindor or Ravenclaw, and Pomona is not in Slytherin. Several complete assignments still satisfy them. For example, Gilderoy Gryffindor, Pomona Hufflepuff, Minerva Ravenclaw, Horace Slytherin works, and so does Gilderoy Ravenclaw, Pomona Gryffindor, and so on. No single symbol is true in all of them.

```
knowledge.add(
    Symbol("MinervaRavenclaw")
)
```

d New clue:

e

```
● amratiah@MacBook-Air-3 CS3081 % /usr/local/bin/python3 /Users/amratiah/CS3081/Labs/Lab2/puzzle.py
GilderoyGryffindor
PomonaHufflepuff
MinervaRavenclaw
HoraceSlytherin
```

f Ravenclaw is now taken, so Gilderoy must be Gryffindor. Pomona can't be Slytherin or Gryffindor or Ravenclaw, so she is Hufflepuff. Horace gets Slytherin. The program prints GilderoyGryffindor, PomonaHufflepuff, MinervaRavenclaw and HoraceSlytherin.

6. Task 3 — Group Challenge: Knights & Knaves (30 marks, 30 min)

On an island, a **knight** always tells the truth and a **knave** always lies. Every person is one or the other, never both. Create a new file `knights.py` that imports `logic.py` and solves the puzzles below.

Hint — use this pattern for what a person says:

<code>Implication(AKnight, statement)</code>	# a knight's statement is true
<code>Implication(AKnave, Not(statement))</code>	# a knave's statement is false

1. **Puzzle 0:** A says “I am both a knight and a knave.”
2. **Puzzle 1:** A says “We are both knaves.” B says nothing.
3. **Puzzle 2:** A says “We are the same kind.” B says “We are of different kinds.”

Bonus (+10 marks) — Puzzle 3: A says either “I am a knight.” or “I am a knave.” (you don’t know which). B says “A said ‘I am a knave.’” B also says “C is a knave.” C says “A is a knight.”

```
1  from logic import *
2
3  AKnight = Symbol("A is a Knight")
4  AKnave = Symbol("A is a Knave")
5  BKnight = Symbol("B is a Knight")
6  BKnave = Symbol("B is a Knave")
7  CKnight = Symbol("C is a Knight")
8  CKnave = Symbol("C is a Knave")
9
10
11 def says(knight, knave, statement):
12     return And(Implication(knight, statement),
13               Implication(knave, Not(statement)))
14
15
16 def kind(knight, knave):
17     return And(Or(knight, knave), Not(And(knight, knave)))
18
19
20 knowledge0 = And(
21     kind(AKnight, AKnave),
22     says(AKnight, AKnave, And(AKnight, AKnave))
23 )
24
25 knowledge1 = And(
26     kind(AKnight, AKnave), kind(BKnight, BKnave),
27     says(AKnight, AKnave, And(AKnave, BKnave))
28 )
29
30 same = Or(And(AKnight, BKnight), And(AKnave, BKnave))
31 different = Or(And(AKnight, BKnave), And(AKnave, BKnight))
32 knowledge2 = And(
33     kind(AKnight, AKnave), kind(BKnight, BKnave),
34     says(AKnight, AKnave, same),
35     says(BKnight, BKnave, different)
36 )
37
38
39 def main():
40     symbols = [AKnight, AKnave, BKnight, BKnave, CKnight, CKnave]
41     puzzles = [
42         ("Puzzle 0", knowledge0),
43         ("Puzzle 1", knowledge1),
44         ("Puzzle 2", knowledge2),
45     ]
46
47     for name, knowledge in puzzles:
48         print(name)
49         for symbol in symbols:
50             if model_check(knowledge, symbol):
51                 print(f"    {symbol}")
52
53
```

- Puzzle 0: A is a Knave.
 - If A were a knight, the statement "I am both" would have to be true, but nobody is both.
 - So A is a knave and the statement is false, which is consistent.
- Puzzle 1: A is a Knave, B is a Knight.
 - If A were a knight, "we are both knaves" would be true, which contradicts A being a knight.
 - So A is a knave and the statement is false. A is a knave, so "both knaves" fails only because B is not a knave. B is a knight.
- Puzzle 2: A is a Knave, B is a Knight.
 - If A were a knight, he and B would be the same kind, so B would be a knight too. Then B's claim "different kinds" would be false, which a knight can't say.
 - So A is a knave and "same kind" is false, so they differ and B is a knight. B's claim "different kinds" is true, which fits.

```

amratiah@MacBook-Air-3 CS3081 % /usr/local/bin/python3 /Users/amratiah/CS3081/Labs/Lab2/knights.py
Puzzle 0
  A is a Knave
Puzzle 1
  A is a Knave
  B is a Knight
Puzzle 2
  A is a Knave
  B is a Knight

```

7. Submission

Create **one public GitHub repository** named `Lab3_GroupX` and submit **only its link** on Blackboard. The repository must contain:

1. `Lab3_GroupX.pdf` — group number, member names, and for each task: your answer, your code, and a screenshot of the output.
2. All `.py` files — `logic.py`, your changed `harry.py` and `puzzle.py`, and your new `knights.py`.
3. `README.md` — member names and one line on how to run `knights.py`.

Before you submit: open your link in a private (incognito) browser window to check that it works.

8. Grading Rubric

Task	What is checked	Marks
Task 1	Correct truth table and clear explanations	30
Task 2	Predictions, results and explanations; a working new clue	30
Task 3	Puzzles 0–2 encoded correctly with output and explanation	30
Submission	Clear PDF and complete GitHub repository	10
Total		100
Bonus	Puzzle 3 solved and explained	+10